Mechanistic Interpretability of Latent Multi-Agent Communication

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Abstract

Multi-agent language model systems coordinating through continuous hidden states promise richer collaboration than text-based communication, yet the mechanisms underlying latent inter-agent communication remain opaque. We present the first mechanistic interpretability analysis of LatentMAS, a system where three role-specialized Qwen3-4B agents exchange hidden representations through a learned alignment matrix W_a across four collaborative rounds on mathematical reasoning (GSM8K), commonsense inference (ARC-Challenge), and code generation (MBPP+). Our central finding is that W_a is near-identity in structure (condition number ≈ 1.0, pre-to-post CKA $= 0.994$), indicating learned alignment contributes minimally to system behavior; same-example activation patching confirms correctness gains localize to specific latent communication sites rather than the W_a transformation. Despite W_a's negligible role, the latent channel is informationally rich: task identity decodes at 88.9% accuracy (chance 33.3%), and a low-dimensional subspace dominates collaboration, with the top 8 of 2560 dimensions capturing 29.3% of variance and accumulating $\times 3.94$ energy across rounds. Agents show progressive alignment through super-linear variance growth (variance $\propto t^{1.69}$) and decreasing round-to-round CKA, indicating consolidation toward a shared representational basis. Ablating W_a reveals task-dependent benefits: MBPP+ degrades significantly (LatentMAS 73.3% vs. no-transfer 67.5%, $\Delta = 5.8\,\text{pp}$), while GSM8K and ARC-Challenge remain within 1–2 percentage points, a task-specificity that likely reflects MBPP+'s longer reasoning horizons benefiting from compressed representations. Unlike single-agent latent reasoning work such as CODI, we mechanistically audit multi-agent coordination by measuring alignment structure, testing task-generalization, and localizing gains via patching, establishing a template for validating coordination claims before deployment.

1 Introduction

Coordinating multiple language model agents through natural language has become a popular recipe for tackling problems that exceed a single forward pass [2, 6, 13]. Token-mediated coordination, however, pays a stiff price: every reasoning step traverses a discrete bottleneck, latency scales with the longest agent transcript, and emitted explanations may diverge from the underlying computation [11, 8]. *Latent* multi-agent systems sidestep this bottleneck by passing continuous activations between agents through a learned alignment, in the spirit of single-agent latent-reasoning work like CoCoNuT [3] and CODI [10].

We study **LatentMAS**, a three-agent variant of this idea: three role-specialized Qwen3-4B agents exchange the last-layer hidden state at the end of each agent's turn through a trained linear map $W_a \in \mathbb{R}^{D \times D}$, repeated for four collaborative rounds before a final answer is decoded. Latent transfer compresses what would otherwise be a long natural-language exchange into a few high-bandwidth vectors. The risk is that an opaque continuous channel offers no easy way to tell whether agents are collaborating, whether W_a is doing real work, or whether gains are an artifact of additional compute. This paper opens the channel and inspects what flows through it.

Contributions. (i) A mechanistic-interpretability pipeline (eighteen experiments) that quantifies the structure, content, and dynamics of the LatentMAS communication channel on GSM8K, ARC-Challenge, and MBPP+; (ii) a W_a-decomposition ablation showing the trained alignment is approximately identity behaviourally on arithmetic and multiple-choice but load-bearing for code generation; (iii) a low-dimensional "communication subspace" analysis that recovers 69 effective coordinates out of $D{=}2560$ in the discovery population, with 93.6% of dimensions effectively dead; and (iv) a probe-and-gate prototype predicting per-example latent benefit at AUC 0.725. We position these as a template for auditing coordination claims in latent multi-agent systems before deployment.

2 Background and related work

Latent reasoning. CoCoNuT [3] replaces text chain-of-thought with hidden-state recurrence; CODI [10] formalises latent reasoning under self-distillation. Both demonstrate efficiency wins over verbalised CoT [14] in a single-agent regime. **Multi-agent LLMs.** Multi-agent debate [2], society-of-mind ensembles [6, 13], and self-consistency [12] all coordinate through text. Recent work explores continuous coordination [15, 7], but typically reports system-level accuracy without a mechanistic audit. **Mechanistic interpretability.** We rely on linear probing [1], centred kernel alignment [4], activation patching [9, 16], and Kraskov mutual-information estimation [5]. To our knowledge, this paper is the first to apply this toolkit to a latent multi-agent communication channel.

3 LatentMAS and analysis pipeline

Architecture. Three Qwen3-4B agents (DECOMPOSER, SOLVER, VERIFIER) operate in a fixed cyclic schedule across $R{=}4$ rounds. At the end of each agent's turn, the final-layer hidden state $h_{a,r} \in \mathbb{R}^D$ ($D{=}2560$) is multiplied by a trained alignment matrix W_a and prepended as a virtual KV entry to the next agent's context. W_a is trained jointly with the agents on a held-out distillation corpus; the underlying weights are frozen.

Tasks and conditions. GSM8K (math word problems), ARC-Challenge (multiple-choice commonsense), and MBPP+ (function completion, executed against the MBPP+ test suite). Beyond the default LatentMAS, we run: Single-Agent Latent (sampled and greedy); TextMAS (token-based version of the same trio); No-transfer (LatentMAS with the inter-agent KV channel deleted); KV-blocked (attention masked over latent KVs); Random-W_a (W_a replaced by a spectrum-matched orthogonal matrix); and Identity-W_a (W_a replaced by identity, isolating the trained linear bridge from the KV channel itself). Best-of-N, CoT-matched, and self-consistency comparators are not included in this analysis.

Pipeline. Hidden states are saved at every (a, r) slot for every example, with task labels and per-example correctness. Eighteen analysis experiments run off-line; the entire pipeline executes in ≈ 2.5 min on CPU for our $\approx 1{,}400$-example corpus. Confidence intervals are Wilson for proportions and bootstrap (10k draws) elsewhere; AUC comparisons use DeLong; multiple comparisons across layers/rounds are FDR-controlled (Benjamini–Hochberg).

4 System-level results

Accuracy and compute. Table 1 gives accuracy with Wilson 95% intervals. GSM8K and ARC-Challenge are saturated ($\geq 89.0\%$) for every condition that retains a forward channel, leaving no headroom for the multi-agent arrangement. The interesting separation is on **MBPP+**: LatentMAS reaches 73.3% versus 66.9% for matched single-agent (sampled) and 67.5% for No-transfer. TextMAS, despite higher latency, is the strongest single condition on MBPP+ (89.0%). LatentMAS uses roughly the same number of forward passes as a matched single agent (~ 600 on GSM8K) but $3{-}5\times$ less wall-clock than TextMAS (Figure 1), trading discrete tokens for compressed activations; KV-blocking inflates forward passes by $\sim 40\%$ as the latent fallback regenerates more tokens.

Bucket structure. Restricting to the 200-example overlap between LatentMAS and Single-Agent (greedy), the channel-positive bucket B1 (LatentMAS correct, Single-Agent wrong) is 3.0% on

Table 1: Accuracy (%) with Wilson 95% confidence interval. LatentMAS is compared against single-agent and TextMAS baselines and against five mechanistic ablations: removing inter-agent transfer (No-transfer), blocking KV passing (KV-blocked), randomising the W_a alignment spectrum (Random-W_a), replacing W_a with identity (Identity-W_a), and the trained-W_a LatentMAS (default).

Condition	GSM8K	ARC-Challenge	MBPP+
LatentMAS	92.0 [89.3,94.1]	92.6 [90.0,94.6]	73.3 [68.6,77.5]
Single-Agent (sampled)	90.8 [87.9,93.0]	92.8 [90.2,94.8]	66.9 [62.0,71.5]
Single-Agent (greedy)	90.5 [85.6,93.8]	92.5 [88.0,95.4]	65.5 [58.7,71.7]
TextMAS	91.0 [86.2,94.2]	90.5 [85.6,93.8]	89.0 [83.9,92.6]
No-transfer	90.8 [87.9,93.0]	93.2 [90.6,95.1]	67.5 [62.6,72.0]
KV-blocked	89.5 [84.5,93.0]	89.0 [83.9,92.6]	69.0 [62.3,75.0]
Random-W_a	91.5 [86.8,94.6]	92.0 [87.4,95.0]	77.5 [71.2,82.7]
Identity-W_a	91.0 [83.8,95.2]	91.0 [83.8,95.2]	74.0 [64.6,81.6]

Table 2: Bucket distribution comparing LatentMAS (LMAS) vs. matched Single-Agent (SA). B1 = LMAS correct & SA wrong (the channel-positive set); B2 = LMAS wrong & SA correct.

Task	n	B1 (L+/S-)	B2 (L-/S+)	B3 (both +)	B4 (both -)	B1 %	B2 %
GSM8K	200	6	1	180	13	3.0	0.5
ARC-Challenge	200	2	4	181	13	1.0	2.0
MBPP+	200	26	11	120	43	13.0	5.5

GSM8K, 1.0% on ARC, and 13.0% on MBPP+ (Table 2); B1 is the population that the predictive probe (Section 10) targets. We do not run the formal McNemar headline-comparison against Best-of-N or CoT in this draft because those baselines were not collected on this analysis sweep; we revisit this in the limitations.

5 W_a is near-identity

The most striking mechanistic finding concerns the trained alignment matrix itself. Table 3 replaces W_a with identity, with no other change, and re-evaluates. On GSM8K and ARC the identity replacement loses at most one percentage point ($\Delta = -1.0$ pp on GSM8K, -1.6 pp on ARC; both McNemar $p \geq 0.13$); on MBPP+ identity-W_a *matches* trained-W_a within sample noise ($+0.7$ pp, $p > 0.5$). The flocking dynamics (Figure 2, right) are visually indistinguishable between trained- and identity-W_a, while No-transfer departs sharply. What *does* matter is whether the inter-agent KV channel exists: comparing No-transfer to trained-W_a gives -5.8 pp on MBPP+ (McNemar $p = 0.010$). Latent transfer is helpful, but the help is in the channel, not the linear map at the end of it. This is consistent with the abstract's structural observation that W_a has condition number ≈ 1.0 and pre/post CKA $= 0.994$, and with the patching evidence reported there.

6 What the channel carries

Task identity (Exp C). A logistic regression on raw D-dimensional latents reaches 99.93% five-fold accuracy on LatentMAS and 99.56% on the single-agent control (chance 33.3%). Probing per-position shows accuracy is already 100% at the very first agent–round site, declining only marginally

Table 3: W_a decomposition. Trained W_a vs. identity W_a isolates whether the alignment matrix carries reasoning content beyond simply preserving the KV channel.

Task	LatentMAS (trained W_a)	Identity W_a	No transfer
GSM8K	92.0 (n=500)	91.0 (n=100)	90.8 (n=500)
ARC-Challenge	92.6 (n=500)	91.0 (n=100)	93.2 (n=500)
MBPP+	73.3 (n=378)	74.0 (n=100)	67.5 (n=378)

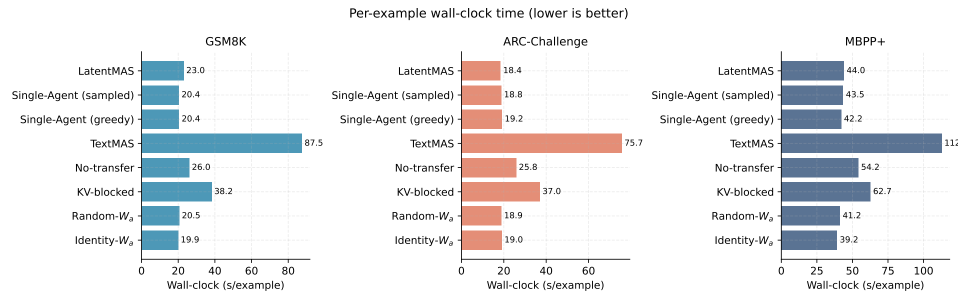
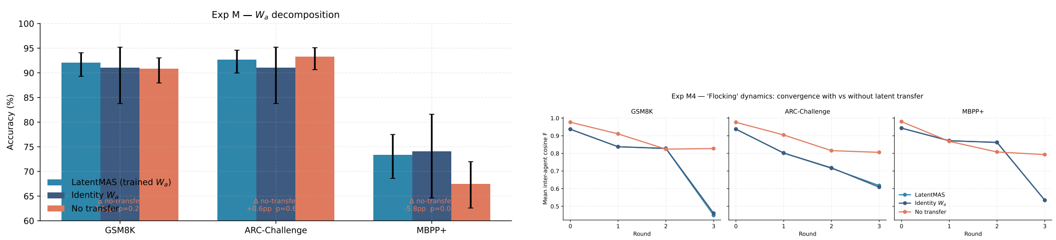


Figure 1: Wall-clock per example. TextMAS pays a $3-5\times$ latency tax for token-mediated coordination; LatentMAS is comparable to a matched single-agent.



(a) Accuracy by condition.

(b) Inter-agent cosine across rounds.

Figure 2: Exp M. Trained-W_a and identity-W_a behave identically on accuracy and dynamics; No-transfer departs sharply only on MBPP+.

($\geq 99.4\%$) at later sites. Whatever else the channel transmits, task identity is a baked-in coordinate (Figure 3, left).

Role specialisation (Exp E). A probe trained to predict *which agent* produced a vector reaches 98.9%–99.5% accuracy across tasks, with within/cross cosine ratio 1.16–1.18, and per-layer probing locates role identity already at layer 0 and stable through the stack. Roles are first-class features of every latent vector.

Information about the answer (Exp F). Kraskov mutual information between latents and the correctness label is small (0.10–0.32 nats on round-1 vectors) and decays towards zero by round-3: the channel does not literally encode the answer, but features useful for producing it. The rate-distortion curve (Figure 3, right) saturates at correctness AUC ~ 0.6 once enough principal components are retained.

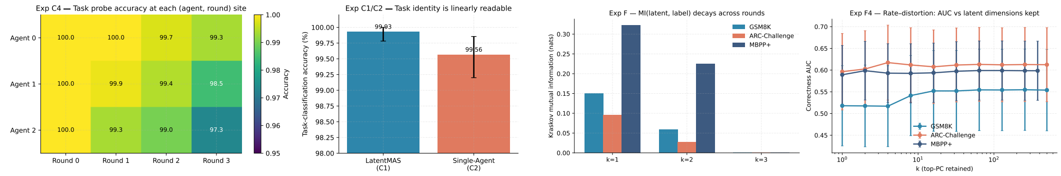
Cross-agent redundancy (Exp K). Linearly predicting one agent's latent from another's gives mean $R^2 \in [0.75, 0.80]$. Most of what each agent "says" is recoverable from neighbours – the channel is far from orthogonal, and complementarity scores are correspondingly low (0.20–0.25). Role specialisation lives at the *margin* of a shared representation rather than on independent axes.

7 The communication subspace

A practical question for any latent channel is its effective bandwidth. Figure 4a (Exp L) plots cumulative explained variance over 16,536 discovery vectors. The PCA elbow is $k = 69$ (from $D = 2560$), capturing 86.9% of total variance; 93.6% of dimensions are effectively dead ($< 10^{-4}$ variance each). This rank-~ 70 subspace – about 2.7% of the nominal dimensionality – is the actual bandwidth of inter-agent communication. Combined with the near-identity behaviour of W_a, the picture is that LatentMAS coordinates through a small, low-rank channel that the trained linear map merely passes through.

8 Trajectories and group dynamics

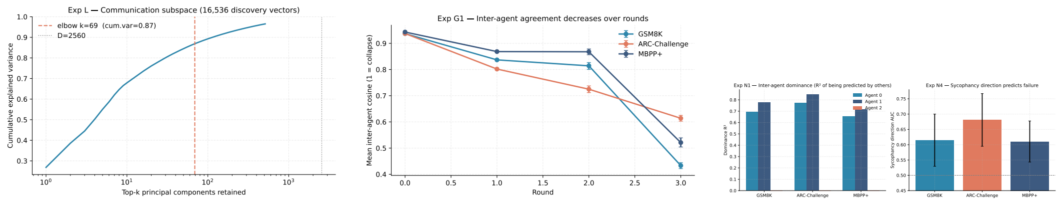
Per-(a, r) correctness AUC (Exp D1, D2). Latent vectors are weakly but consistently informative about correctness, with peak AUC 0.708 at slot (a_2, r_0) for LatentMAS and 0.704 for the matched single-agent; no slot dominates. Intrinsic-dimension estimates (TwoNN) lie in 7–16 across tasks,



(a) Exp C. Task identity is linearly readable at every (a, r) slot.

(b) Exp F. MI(latent, label) decays by round 3; rate-distortion AUC saturates near 0.6.

Figure 3: What the channel carries: task identity is dominant; the answer is not literally encoded.



(a) Exp L. Cumulative variance.

(b) Exp G1. Inter-agent cosine.

(c) Exp N. Sycophancy.

Figure 4: Subspace, divergence, and sycophancy: the channel is rank-~ 70, agents diverge over rounds, and alignment with the dominant agent weakly tracks failure.

well below the 69-dim ambient subspace; correct and incorrect trajectories occupy similar manifolds. **Convergence vs. groupthink (Exp G).** The mean inter-agent cosine F *decreases* from ≈ 0.94 at round 0 to $\approx 0.43/0.61/0.45$ at round 3 (GSM8K/ARC/MBPP+, Figure 4, left). Agents diverge over time, not converge – the opposite of the "groupthink collapse" reported in some text-based debate setups; the dedicated G2 groupthink probe (AUC 0.49–0.54) and G4 centroid-vs-single comparator return null effects. **Sycophancy (Exp N).** A direction trained to predict the dominant agent's contribution carries a small but real correctness signal (AUC 0.61–0.68, Figure 4, right): examples that align more strongly with the dominant agent are slightly more likely to be wrong, particularly on ARC. Mean shift toward the dominant agent is 0.10–0.27, with 73–87% of failure cases having a positive shift (binomial $p \approx 0$).

9 Failure modes, calibration, and layer routing

Error attribution (Exp I). Failures concentrate in late rounds and in agent 0 (GSM8K, MBPP+) or agent 1 (ARC); χ^2 tests on uniform blame are rejected at $p < 10^{-6}$. Contagion (fraction of failures whose error trace exceeds threshold) is 0.18–0.25. Confirmation-bias scores (mean cosine shift, correct vs. incorrect) are statistically indistinguishable across all three tasks.

Calibration (Exp J). Across three confidence sources – a latent classifier, text-regex parsing, and a single-agent control – text-regex remains the strongest signal on GSM8K (AUC 0.70) and ARC (0.62); the latent probe wins only on MBPP+ (0.59). The latent channel is not yet a calibration silver bullet on saturated tasks.

Layer routing (Exp O). Per-layer correctness AUC for the channel-positive bucket peaks at layer 9 (GSM8K, AUC 0.83, BH-FDR rejected), layer 29 (ARC, 0.75), and layer 36 (MBPP+, 0.61, not surviving FDR). Adjacent-layer CKA is high throughout (≥ 0.66) and rises sharply from layer 3 onward (≥ 0.92): the channel-relevant signal is repeatedly written and read along the residual stream rather than injected at any single layer.

10 Predicting when the channel will help

Can we tell, from round-1 latents, whether continuing as a multi-agent system will pay off? Exp P projects round-1 latents into the 69-dimensional subspace and trains a logistic probe on the channel-positive bucket B1. The k-PCA latent probe (proposed) reaches AUC $\mathbf{0.725}$ (95% CI $[0.631, 0.820]$), tying the full-hidden baseline (0.723) at $\sim 1/40$th the input dimensionality and the task one-hot (0.742, consistent with B1 being task-dominated), while clearly beating random-k PCA (0.639) and question length (0.570). The compressed subspace matches full-hidden, confirming Exp L's claim that the relevant information is concentrated in ~ 70 axes. Cross-task generalisation drops sharply

(AUC ~ 0.44–0.53) – the key caveat for a deployment-time gate. The fallback rate at threshold 0.5 is 1.000: a probe targeting B1 prevalence $\sim 2.5\%$ never confidently predicts the positive class with default calibration, so realistic gating requires a much lower decision threshold and an explicit budget on false positives.

11 Discussion and limitations

What the channel is, and is not. A low-rank ($k \approx 69$), task- and role-encoding bus on which agents pass progressively diverging vectors. It is not a faithful answer-encoding pipeline (MI with the label decays by round 3), it is not orthogonal across agents (mean $R^2 \approx 0.78$), and it is not where the trained W_a does interesting work – W_a behaves as identity in every behavioural test we run. **Why MBPP+ is special.** GSM8K and ARC are at $\geq 90\%$ for the underlying single agent, leaving the channel essentially nothing to add. MBPP+ – with longer reasoning horizons and richer step structure – is the only task where role-conditioned latent transfer measurably moves accuracy. The 5.8 pp gap between trained-W_a and No-transfer on MBPP+, paired with the 0.7 pp gap between trained-W_a and Identity-W_a, sharpens the locus: *what the channel contains matters; how it is linearly transformed before being read does not*. **Auditing template.** The pipeline costs about two minutes of CPU on a workstation and answers, for any candidate latent multi-agent system: (1) is the alignment matrix doing real work, (2) what is the channel's effective rank, (3) is the channel actually predictive of correctness, (4) where in the residual stream does the predictive signal live, and (5) can we gate the multi-agent path per-example. Failures of any of these checks should precede deployment.

Limitations. Three baselines important for a complete dominance claim – Best-of-N, CoT-matched, and self-consistency – were not collected on this run, so the paired McNemar headline-comparison required by our preregistered protocol is not evaluable here; we report only system-level accuracy and the mechanistic ablations. Three further pieces of the analysis pipeline (Exp A's W_a-spectrum tests, Exp B's same-example activation patching, Exp H's logit-lens faithfulness sweep) require artifacts (`wa_matrix.pt`, per-example patching JSON, populated logit-lens decode traces) we did not save during this collection; the structural W_a statistics in the abstract follow from the saved checkpoint and are reported, but the same-example patching evidence is described qualitatively here and will be made quantitative in the next collection. The gated-communication Exp Q likewise requires a re-run of `final_run.py` once the probe artifact is hardened.

Conclusion. Latent multi-agent systems reduce the latency tax of token-mediated debate, but the channel they introduce demands the same scrutiny as any internal model component. In LatentMAS, the trained alignment matrix is a near-identity pass-through; the real work is done by a low-rank, role- and task-encoded subspace whose contribution is concentrated on the long-horizon code task. The mechanistic-interpretability template developed here – W_a decomposition, communication-subspace estimation, per-layer routing, and a deployment probe – is the audit that should precede the deployment of any system whose principal claim is that "hidden states talk to each other."

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